

Bijal Panchal

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SUMMARY

Turning fragmented, multi-source data into decisions is where I thrive — as a Data & BI Professional (M.S. Data Analytics, Distinction) with hands-on experience across warehouse design, predictive modeling, and business reporting. I've built star-schema warehouses to unify scattered datasets, automated ETL pipelines in Python and SQL, and developed predictive models — reducing data inconsistencies by 30%, boosting query performance by 40%, and reaching up to 0.89 AUC-ROC in model accuracy. Skilled across Python, SQL, Power BI, and Tableau, with a knack for translating technical findings into narratives business stakeholders can act on. Proficient in Microsoft Office (Word, Excel, PowerPoint), with ongoing expansion into Azure and Databricks. Available immediately for full-time roles.

SKILLS

Languages: Python (Pandas, NumPy, Scikit-learn, Statsmodels, Matplotlib, Seaborn), SQL, VBA, SQLAlchemy

Modeling & Analysis: K-Means Clustering, XGBoost, PCA, ARIMA, SARIMA, Holt-Winters, Prophet, Isolation Forest, Feature Engineering, Model Interpretability, Model Evaluation & Validation (Precision, Recall, F1-Score, AUC-ROC), Train-Test Validation

Data Governance & Compliance: Standardized Workflow & Process Development, Data Quality Management, Documentation & Traceability, Root Cause Analysis, Reconciliation

BI/Visualization: Tableau, Power BI (DAX, Power Query), Streamlit, Excel, PowerPoint, Word

Data & Warehousing: Structured & Unstructured Data, MySQL, SQLite, PostgreSQL, Star Schema Design, ETL Pipelines, Dimensional Modelling, API Integration

DevTools: Git, GitHub, GitHub Actions, Jupyter Notebook, VS Code, pytest

CloudStack: Azure (developing), Databricks (developing), Snowflake, dbt, GCP BigQuery, AWS S3/Redshift

WORK EXPERIENCE

POINTS AFRICA

Aug 2025 - Feb 2026

Data Analyst Intern

Toronto, ON (Remote)

- Built a PostgreSQL star-schema warehouse unifying data from 3 enterprise partners (GLICO, JUMIA, STAR) using Python and SQL, reducing data inconsistencies by **30%** and improving query performance by **40%**.
- Cleaned and standardized transactional, loyalty, and geographic datasets at scale, resolving formatting errors and duplicates to achieve **100% field coverage** across all partner sources.
- Applied RFM segmentation to classify customer loyalty tiers, identifying high-value clusters across all 3 partner networks to inform targeted engagement strategies.
- Delivered **10+ Tableau dashboards** tracking KPIs, partner performance, and regional trends — directly adopted by senior stakeholders for data-driven decision-making.

DMG GROUP OF COMPANIES

May 2023 - Sep 2024

Data Analyst

Ahmedabad

- Architected a PostgreSQL star-schema data warehouse unifying enrollment, LMS activity, and assessment data for **100,000+ learners** across IT certification programs, writing complex **SQL** queries for extraction, transformation, and validation to establish a single source of truth for academic reporting.
- Built and validated a dropout-risk prediction model (XGBoost) achieving **0.89 AUC-ROC** and **0.82 F1-score**, enabling academic teams to identify at-risk students early and prioritize intervention before program dropout occurred.
- Designed **Power BI** dashboards and **Excel**-based reporting to surface high-risk student cohorts, enrollment trends, and model performance metrics, giving non-technical academic staff visibility without needing to query the underlying data.

- Delivered a complementary real-time Streamlit dashboard as an early-warning tool for high-risk cohorts, and implemented automated model monitoring (drift detection via PSI/KS-test) to keep predictions reliable as new data flowed in.

PROJECTS

City Traffic Intelligence Platform | Python, SQLite, Streamlit, Scikit-learn, Prophet — Live

- Engineered a **fully automated real-time data platform** ingesting daily metrics across **5 global cities** from **5 live APIs** (TomTom, OpenAQ, Open-Meteo, Numbeo, City Open Data) — built on modular Python collectors feeding a SQLite star schema warehouse (**9 tables, 4,400+ rows**) via a self-running GitHub Actions pipeline.
- Designed a **weighted City Stress Index scoring engine** (Traffic 35% + Air Quality 25% + Weather 20% + Safety 10% + Cost of Living 10%) powered by **15+ engineered features** — including rolling averages, volatility scores, and heat stress index — using ARIMA, Prophet, Z-Score, Isolation Forest, and K-Means models.
- Deployed a **7-page interactive Streamlit dashboard** (live at city-stress-index.streamlit.app) with real-time stress scores, 30-day forecasts, anomaly detection alerts, and cross-city clustering — validated by a **22-test pytest suite** ensuring pipeline and database integrity.

Retail Sales Forecasting & Demand Planning | Python, SQL, Power BI, Statsmodels, SQLite

- Cleaned and processed **1M+ transactional records** across **43 countries and 4,735 products**, resolving 34K duplicates, 19K cancellations, and 243K missing customer IDs — then engineered a SQL star schema warehouse (1 fact + 4 dimension tables) via a custom Python ETL pipeline (SQLAlchemy).
- Built and benchmarked **SARIMA and Holt-Winters** forecasting models on monthly revenue data, with SARIMA achieving **74.5% accuracy** (MAPE 25.54%) on a chronological 80/20 train-test split — reducing simulated stockout rates by **18%**.
- Delivered a **4-page Power BI dashboard** with dynamic KPI cards, seasonal trend lines, and forecast-vs-actual comparisons (80% confidence intervals) — powering RFM segmentation across 6 customer tiers that identified **843 Champions** (avg. \$5,689 spend) and flagged 409 At-Risk customers as the top re-engagement priority.

EDUCATION

University of Niagara Falls

OCT 2024 - APR 2026

Data Analytics

- Graduated with Distinction, GPA 3.89/4.0 — President's Honour Roll, 100% in Advanced Analytics Internship.

Navrachana University

JUL 2020 - JUN 2024

Computer Science & Engineering

- Coursework in Data Structures, DBMS, AI, Cloud Computing, and Cybersecurity; completed a 12-credit Industry Internship and Major Project.

CERTIFICATIONS & ACTIVITIES

Advanced Power BI: DAX Language, Formulas, and Calculations | Course Completion — Apr 2025

- A targeted Power BI certification covering core skills in data modeling, DAX language, and advanced DAX formulas. Emphasizes practical application through dynamic dashboards and real-world business intelligence scenarios.

Hack For Change | Participant — Apr 2022

- Participated in a 49-hour hackathon organized by Navrachana University in collaboration with NuvYuva. Collaborated with peers to develop innovative solutions aimed at social impact and technology-driven change. Strengthened skills in project development, teamwork, and rapid problem-solving under time constraints.